

Strong Pip Algorithm

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We start with a full infinite length tangle W , where W^U and W^S are the unstable and stable manifolds, respectively. $W^U(z_i, q_i)$ is the open interval of W^U between the two points $z_i, q_i \in W^U$. Additionally, $\mathcal{M}(q_i) = q_{i+1}$. We are interested in classifying whether a given intersection point is a strong pip. A point q_0 is a *strong pip* if it satisfies the following condition.

$$\left[\bigcup_{i=0}^{n-1} W^U(z_i, q_i) \right] \cap \left[\bigcup_{i=0}^{n-1} W^S(z'_i, q_i) \right] = \emptyset.$$

That is, there are no intersections between the stable and unstable manifolds up to the points q_i . To determine whether a point is a strong pip in practice, we must utilize a finite set of intersection points called a trellis.

Begin with a trellis T connected to a finite set of periodic points $z_i \in P$ and $z'_i \in P'$ with period n . We are concerned with classifying the intersections of the trellis $N = T^U \cap T^S$. We then restrict ourselves to only looking along the branch of the stable manifold containing q_0 . Any intersections on the stable branches between z'_i and q_i where $i > 0$ would occur first on the stable branch attached to z_0 , so it is sufficient to check just this interval as follows:

$$\left[\bigcup_{i=0}^{n-1} W^U(z_i, q_i) \right] \cap W^S(z'_0, q_0) = \emptyset.$$

So, no intersections can lie between the periodic point z'_0 and intersection point q_0 in order for q_0 to be a strong pip.

To guarantee a trellis contains enough information to determine if a point is a strong pip we must have a point and its $(n-1)^{th}$ iterate. That condition forces any intersections that would disqualify q_0 from being a strong pip to be computed already. This can be any point since we will look at all intersections in the trellis to determine if a given point is a strong pip or not.

Now, take every intersection point and map it back to the stable branch that is attached to z'_0 . We compare the canonical distance of each of these intersections to our candidate point. If there exists no canonical distance less than that of our candidate point, then the candidate is a strong pip.

Note that to do this backward mapping it is not necessary to use the actual dynamical map, but rather the property of canonical distances. That is that

$$c_{dist}^S(\mathcal{M}^{(-i)}(r_0)) = \frac{c_{dist}^S(r_0)}{\alpha^i}.$$

Where $c_{dist}^S(r_0)$ is the stable canonical distance at point r_0 and $\mathcal{M}^{-i}$ is the inverse map applied i times. So to map all intersections back onto the proper stable branch one only needs to do simple division rather than utilize the full dynamical map.

Algorithm 1 IS-STRONG-PIP(q_0, T) - Returns if q_0 is a strong pip or not

- 1: Let N be the set of all intersections in the trellis
 - 2: Take each point in N and map it backwards the minimum number of times required to get it onto $W^S(z'_0, q_0)$ to form $\bar{N}$.
 - 3: **if** there exists a point $r \in \bar{N}$ with a unstable canonical distance less than that of q_i **then**
 - 4: **return** FALSE (q_0 is not a strong pip)
 - 5: **else**
 - 6: **return** TRUE (q_0 is a strong pip)
 - 7: **end if**
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